

Weather Stations Monitoring — Reference Manual

Technical terms and technologies, with sources in reading order

How to use this document

Every technical term and technology used in the Weather Stations Monitoring project is listed below, in the order the project depends on it. Each entry names where the concept appears in the codebase and points to the primary source that documents it. The sources are the explanation; this document only orders them.

All links were checked and resolve at the time of writing.

Contents

1. Streaming foundations
2. Language and build
3. Kafka producer — the weather stations
4. Kafka consumer — the central station
5. Kafka Streams — the rain detector
6. Relational persistence
7. Analytical SQL
8. Containers
9. Kubernetes
10. Local cluster

1. Streaming foundations

Read first. Everything else assumes the vocabulary established here.

Event streaming, brokers, records

Apache Kafka — Introduction

<https://kafka.apache.org/documentation/#introduction>

Topics, partitions, offsets, producers, consumers, consumer groups

Apache Kafka — Main Concepts and Terminology

https://kafka.apache.org/documentation/#intro_concepts

Log structure, replication, delivery semantics (at-most-once, at-least-once, exactly-once)

Apache Kafka — Design

<https://kafka.apache.org/documentation/#design>

Running a broker and producing and consuming a first message

Apache Kafka — Quickstart

<https://kafka.apache.org/quickstart>

Coordination service used by the broker for cluster metadata

Apache ZooKeeper — Overview

<https://zookeeper.apache.org/doc/current/zookeeperOver.html>

Where in the project: `k8s/kafka/kafka.yaml`, `k8s/kafka/zookeeper.yaml`,
`common/src/main/java/com/weathermonitoring/common/util/Topics.java`

2. Language and build

Java 17 — language and standard library

Oracle — Java SE 17 Documentation

<https://docs.oracle.com/en/java/javase/17/>

Project Object Model, dependencies, dependency management

Apache Maven — Introduction to the POM

<https://maven.apache.org/guides/introduction/introduction-to-the-pom.html>

Multi-module (reactor) builds, module ordering, -pl and -am

Apache Maven — Guide to Working with Multiple Modules

<https://maven.apache.org/guides/mini/guide-multiple-modules.html>

Uber-JAR / shaded JAR packaging, manifest main class

Apache Maven — Shade Plugin

<https://maven.apache.org/plugins/maven-shade-plugin/>

JSON serialisation and deserialisation, field binding annotations

FasterXML — Jackson Databind

<https://github.com/FasterXML/jackson-databind>

Logging facade and binding

SLF4J — User Manual

<https://slf4j.org/manual.html>

Where in the project: pom.xml, common/pom.xml, weather-station/pom.xml, central-station/pom.xml,
common/src/main/java/com/weathermonitoring/common/util/JsonUtil.java

3. Kafka producer — the weather stations

Producer API, send, callbacks, partitioning by key

Apache Kafka — KafkaProducer Javadoc (3.7)

<https://kafka.apache.org/37/javadoc/org/apache/kafka/clients/producer/KafkaProducer.html>

acks, enable.idempotence, retries, batch.size, buffer.memory

Apache Kafka — Producer Configs

<https://kafka.apache.org/documentation/#producerconfigs>

Where in the project:

weather-station/src/main/java/com/weathermonitoring/station/WeatherStationApp.java

4. Kafka consumer — the central station

Consumer API, poll loop, subscription, rebalancing

Apache Kafka — KafkaConsumer Javadoc (3.7)

<https://kafka.apache.org/37/javadoc/org/apache/kafka/clients/consumer/KafkaConsumer.html>

group.id, auto.offset.reset, enable.auto.commit, manual commitSync

Apache Kafka — Consumer Configs

<https://kafka.apache.org/documentation/#consumerconfigs>

Where in the project:

central-station/src/main/java/com/weathermonitoring/central/CentralStationApp.java

5. Kafka Streams — the rain detector

Stream processing topology, KStream, stream threads, application id

Apache Kafka — Streams Core Concepts (3.7)

<https://kafka.apache.org/37/documentation/streams/core-concepts>

Streams Developer Guide

<https://kafka.apache.org/37/documentation/streams/developer-guide/>

DSL operators: mapValues, filter, to; stateless transformations

Apache Kafka — Streams DSL API

<https://kafka.apache.org/documentation/streams/developer-guide/dsl-api.html>

Where in the project:

`central-station/src/main/java/com/weathermonitoring/central/RainDetectorApp.java`

6. Relational persistence

JDBC connections, statements, result sets

Oracle — JDBC Basics

<https://docs.oracle.com/javase/tutorial/jdbc/basics/index.html>

PreparedStatement, parameter binding, addBatch / executeBatch

Oracle — Using Prepared Statements

<https://docs.oracle.com/javase/tutorial/jdbc/basics/prepared.html>

PostgreSQL JDBC driver, connection URLs, TLS parameters

pgJDBC — Documentation

<https://jdbc.postgresql.org/documentation/>

CREATE TABLE, column constraints, table constraints

PostgreSQL 16 — CREATE TABLE

<https://www.postgresql.org/docs/16/sql-createtable.html>

BIGSERIAL and the serial pseudo-types

PostgreSQL 16 — Numeric Types

<https://www.postgresql.org/docs/16/datatype-numeric.html>

Unique constraints and their backing indexes

PostgreSQL 16 — Unique Indexes

<https://www.postgresql.org/docs/16/indexes-unique.html>

INSERT ... ON CONFLICT DO NOTHING (upsert, idempotent insert)

PostgreSQL 16 — INSERT

<https://www.postgresql.org/docs/16/sql-insert.html>

Transactions, commit and rollback

PostgreSQL 16 — Transactions

<https://www.postgresql.org/docs/16/tutorial-transactions.html>

Where in the project: db/schema.sql,

central-station/src/main/java/com/weathermonitoring/central/WeatherReadingRepository.java,

central-station/src/main/java/com/weathermonitoring/central/DbConfig.java

7. Analytical SQL

SELECT, GROUP BY, aggregate functions

PostgreSQL 16 — SELECT

<https://www.postgresql.org/docs/16/sql-select.html>

Window functions and OVER (PARTITION BY ...)

PostgreSQL 16 — Window Functions

<https://www.postgresql.org/docs/16/tutorial-window.html>

Where in the project: db/analysis_queries.sql

8. Containers

Dockerfile instructions: FROM, COPY, RUN, ENV, ENTRYPOINT

Docker — Dockerfile Reference

<https://docs.docker.com/reference/dockerfile/>

Multi-stage builds, build stages, copying artefacts between stages

Docker — Multi-stage Builds

<https://docs.docker.com/build/building/multi-stage/>

Image architectures and multi-platform images (amd64, arm64)

Docker — Multi-platform Builds

<https://docs.docker.com/build/building/multi-platform/>

Where in the project: weather-station/Dockerfile, central-station/Dockerfile

9. Kubernetes

Read in this order; each concept builds on the previous.

Pods — the unit of scheduling

<https://kubernetes.io/docs/concepts/workloads/pods/>

Deployments — stateless replica management

<https://kubernetes.io/docs/concepts/workloads/controllers/deployment/>

StatefulSets — stable ordinal identities and hostnames

<https://kubernetes.io/docs/concepts/workloads/controllers/statefulset/>

Services — stable virtual IPs, DNS names, headless services

<https://kubernetes.io/docs/concepts/services-networking/service/>

ConfigMaps — non-secret configuration injected as files or variables

<https://kubernetes.io/docs/concepts/configuration/configmap/>

Secrets — credential storage and injection

<https://kubernetes.io/docs/concepts/configuration/secret/>

PersistentVolumes and PersistentVolumeClaims — durable storage

<https://kubernetes.io/docs/concepts/storage/persistent-volumes/>

Resource requests and limits, scheduling and eviction

<https://kubernetes.io/docs/concepts/configuration/manage-resources-containers/>

Liveness, readiness and startup probes

<https://kubernetes.io/docs/tasks/configure-pod-container/configure-liveness-readiness-startup-probes/>

Where in the project: `k8s/namespace.yaml`, `k8s/weather-stations.yaml`,
`k8s/central-station.yaml`, `k8s/rain-detector.yaml`,
`k8s/database/postgres.yaml`, `k8s/kafka/kafka.yaml`,
`k8s/kafka/zookeeper.yaml`

10. Local cluster

Running a single-node Kubernetes cluster locally

Kubernetes — Hello Minikube tutorial

<https://kubernetes.io/docs/tutorials/hello-minikube/>

minikube — source, releases and command reference

<https://github.com/kubernetes/minikube>

Where in the project: k8s/README.md